Fred Dickinson fredsterdickinson@gmail.com

$\sqrt{p}$ is Irrational

Proof that $\sqrt{p}$ is irrational for all prime numbers p: three cases.

This is a very common question in A-level maths, tackled using **proof by contradiction.** Recall that a number x is rational if it can be expressed as a fraction, $x = \frac{a}{b}$, where a and b are integers and $b \neq 0$. Furthermore, note that if this fraction exists, then it can be written in its simplest form.

Example 1: show that $\sqrt{2}$ is irrational.

1. For a contradiction, assume that $\sqrt{2}$ is rational. Then $\sqrt{2}$ can be expressed $\sqrt{2} = \frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and the fraction is written in its simplest form.
2. Squaring both sides, $2 = \frac{a^2}{b^2}$, and therefore $2b^2 = a^2$.
   (a) Therefore a^2 is a multiple of 2.
   (b) Since a^2 is a multiple of 2, it follows that a is also a multiple of 2.
   (c) Therefore $a = 2m$ for some integer m.
3. Rearranging, $b^2 = \frac{a^2}{2}$. Using $a = 2m$, it follows $b^2 = \frac{4m^2}{2} = 2m^2$.
   (a) Therefore b^2 is a multiple of 2.
   (b) Since b^2 is a multiple of 2, it follows that b is also a multiple of 2.
   (c) Therefore $b = 2n$ for some integer n.
4. So both a and b are multiples of 2 and the fraction $\frac{a}{b}$ is not in its simplest form. we have reached a contradiction, and therefore $\sqrt{2}$ is irrational.

Comments: this is the general framework for any proof that $\sqrt{p}$ is irrational. The most important steps that you *cannot miss* are the hypotheses in step 1.

After that, the 'glue' holding the proof together are the claims in 2(b) and 3(b). Why are these true? (Solution given on pg. 3 - please think about it first).

Overleaf is a second example that proves $\sqrt{7}$ is irrational.

Example 2: show that $\sqrt{7}$ is irrational.

1. For a contradiction, assume that $\sqrt{7}$ is rational. Then $\sqrt{7}$ can be expressed $\sqrt{7} = \frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and the fraction is written in its simplest form.
2. Squaring both sides, $7 = \frac{a^2}{b^2}$, and therefore $7b^2 = a^2$.
   (a) Therefore a^2 is a multiple of 7.
   (b) Since a^2 is a multiple of 7, it follows that a is also a multiple of 7.
   (c) Therefore $a = 7m$ for some integer m.
3. Rearranging, $b^2 = \frac{a^2}{7}$. Using $a = 7m$, it follows $b^2 = \frac{49m^2}{7} = 7m^2$.
   (a) Therefore b^2 is a multiple of 7.
   (b) Since b^2 is a multiple of 7, it follows that b is also a multiple of 7.
   (c) Therefore $b = 7n$ for some integer n.
4. So both a and b are multiples of 7 and the fraction $\frac{a}{b}$ is not in its simplest form. we have reached a contradiction, and therefore $\sqrt{7}$ is irrational.

Comments: you should notice that this looks very similar to the first example. In fact, it's a copy-paste job, just with the 2's replaced with 7's.

The 'glue' of the proof is the same claim that appears in 2(b) and 3(b). Can you prove them again?

HINT: use the prime factorisation theorem, that says that any integer can be expressed as a product of its prime factors. That is,

$$x = p_1^{r_1} p_2^{r_2} \cdots p_N^{r_N},$$

where $p_1, \ldots, p_N$ are distinct prime numbers and $r_1, \ldots, r_N$ are positive integer powers. For an example, consider

$$29250 = 2 \times 3^2 \times 5^3 \times 13.$$

My proof of the claims in 2(b) and 3(b) is given on the next page.

Fred Dickinson fredsterdickinson@gmail.com

Claim: let P be a prime number, and x any integer. If x^2 is a multiple of P, then x is also a multiple of P.

Proof: consider the prime factorisation of x,

$$x = p_1^{r_1} p_2^{r_2} \cdots p_N^{r_N},$$

where $p_1, \ldots, p_N$ are distinct prime numbers and $r_1, \ldots, r_N$ positive integer powers. Then the prime factorisation of x^2 is

$$\begin{aligned} x^2 &= (p_1^{r_1} p_2^{r_2} \cdots p_N^{r_N}) \times (p_1^{r_1} p_2^{r_2} \cdots p_N^{r_N}) \\ &= p_1^{2r_1} p_2^{2r_2} \cdots p_N^{2r_N}. \end{aligned}$$

So if x^2 is a multiple of P, then P exists in this prime factorisation (i.e. P is one of the primes $p_1, \ldots, p_N$). Therefore P is a factor of x too. □

Comments. Sorry for the slightly confusing notation with (capital) P and (lowercase) $p_1, \ldots, p_N$ - it's convenient to use p to indicate prime numbers. You may wish to use $q_1, \ldots, q_N$ instead.

This is a reasonably quick and simple proof. The prime factorisation theorem can be incredibly helpful in a wide number of cases - it's not a bad idea to try and apply it if you are stuck on a similar proof!

An important technical point is that the prime factorisation is **unique**, meaning it is the only possible way for prime numbers to make up x.

To finish these notes, we give the final - and most general - proof on the next page. **Try it yourself first:** prove that if p is a prime number, then $\sqrt{p}$ is irrational.

Fred Dickinson fredsterdickinson@gmail.com

Example 3 (final, most general case): show that $\sqrt{p}$ is irrational for all prime numbers p.

1. For a contradiction, assume that $\sqrt{p}$ is rational. Then $\sqrt{p}$ can be expressed $\sqrt{p} = \frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and the fraction is written in its simplest form.
2. Squaring both sides, $p = \frac{a^2}{b^2}$, and therefore $pb^2 = a^2$.
   (a) Therefore a^2 is a multiple of p.
   (b) Since a^2 is a multiple of p, it follows that a is also a multiple of p.
   (c) Therefore $a = pm$ for some integer m.
3. Rearranging, $b^2 = \frac{a^2}{p}$. Using $a = pm$, it follows $b^2 = \frac{p^2m^2}{p} = pm^2$.
   (a) Therefore b^2 is a multiple of p.
   (b) Since b^2 is a multiple of p, it follows that b is also a multiple of p.
   (c) Therefore $b = pn$ for some integer n.
4. So both a and b are multiples of p and the fraction $\frac{a}{b}$ is not in its simplest form. we have reached a contradiction, and therefore $\sqrt{p}$ is irrational.

Comments. Hopefully this more general case is not too bad - it is, again, almost identical to examples 1 and 2.